

Antonio Carmagnani

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PROFILE

Experience in international infrastructure consulting on water and energy projects across emerging economies, working from tender through delivery. Recently completed a Master's Degree in Architecture Computation at The Royal Danish Academy, with a focus on data-driven approaches to the built environment. Hands-on skills span climate risk assessment (JASPERS/EBRD), GIS-based spatial analysis, and preparing technical documentation and tenders for development-bank-funded projects. Motivated by work where infrastructure, sustainability, and technical analysis intersect.

EXPERIENCE

Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S

Copenhagen, DK

02.2026 - 07.2026

- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.
- Authored climate risk assessments for major water supply projects following JASPERS and EBRD climate-proofing guidance, screening hazards against projections data and rating risk by exposure and vulnerability for each asset.
- Joined a field mission across Baku's water supply system, assessing asset condition and identifying technical, social, environmental, and human rights risks through site inspections and interviews.
- Built geospatial databases for water and infrastructure projects in QGIS, compiling georeferenced asset data, satellite imagery, and open sources for socio-spatial analysis.
- Developed a pricing tool to support bid decisions, parametrizing expert fee bands and project economics for financial viability.

Intern in Responsible Business Conduct @ Ramboll Management Consulting

Copenhagen, DK

05.2025 - 08.2025

- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.
- Developed the firm's real estate ESG due diligence services, centered on building energy and decarbonization assessments aligned with EU building performance regulations.
- Built a Human Rights Impact Assessment tool, mapping 40+ salient human rights risks across supply chains, with severity scoring to prioritize the most material risks.
- Developed a salience-based human rights due diligence, scoring risks across business units and supply chains by probability, scale, and scope to automatically prioritize salient risks.
- Mapped the global supply chains of 20+ critical raw materials under the EU Critical Raw Materials Act, profiling producing companies, sectors, and geographies from authoritative industry databases.

Architect @ Isay Weinfeld Architects

São Paulo, BR

02.2023 - 08.2024

- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.

Trainee in Real Estate Product Development @ OSPA

São Paulo, BR

02.2022 - 10.2022

- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.
- Ran financial feasibility studies on candidate sites, sizing investment and return against buildable area, unit mix, price per square metre, and parking to test viability.
- Produced images, plans, and diagrams and presented feasibility outcomes to developer clients.

Architect @ Felipe Hess Architects

São Paulo, BR

07.2021 - 02.2022

- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.

Intern in Project Management @ Itaú-Unibanco

São Paulo, BR

08.2020 - 07.2021

- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).
- Assessed the environmental and energy performance of buildings the bank was acquiring or leasing, reviewing consumption data, efficiency ratings, and certifications as part of internal due diligence.

Independent Consultant @ Self-employed

São Paulo, BR and Copenhagen, DK

07.2018 - Present

- Took on independent projects, software, and consulting for clients across the AEC sector, alongside studies and full-time roles.
- Co-founded and ran a small practice delivering projects for 10+ clients in the AEC sector, responsible for client acquisition, scoping, pricing, and delivery.
- Advised a software startup in the built environment over two years on product direction, UI and UX, and applied AI, testing releases and returning improved Revit and Dynamo scripts.

EDUCATION

Master's Degree in Architecture Computation @ The Royal Danish Academy

Copenhagen, DK

07.2024 - 07.2026

Semester Programme in Building Technology @ Architecture Association

London, UK

07.2022 - 01.2023

Summer School in Parametric Design @ Architecture Association

London, UK

06.2019 - 07.2019

07.2018 - 12.2023

PROJECTS**Rammed Blue Biomass & Computational Optimization @ The Royal Danish Academy****Copenhagen, DK**

- Developed a new construction technique and material system using blue biomass (seashells and algae), applying LCA metrics and parametric simulations to optimize structural performance and environmental impact.
- Project was selected as a semester highlight and served as the material foundation for the school's installation at the Venice Architecture Biennale 2025, presented in collaboration with CITA, GXN, and 3XN at the Arsenale.

Real Estate Development Recommendation Tool @ The Royal Danish Academy**Copenhagen, DK**

- Built a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by integrating financial, social, sustainability, and geospatial data.
- Designed a user interface allowing different stakeholders to adjust impact drivers and weighting, with predictive outputs aligned to ESG priorities and local development conditions.

Master's Thesis, Computational Urban Analysis for Intensification Strategy @ The Royal Danish Academy**Copenhagen, DK**

- Developed a computational framework to identify underused spaces in the historic city center of São Paulo, Brazil, combining urban morphology analysis, land use data, and social vulnerability indicators.
- Built an algorithm that uses urban data, including building typology, use patterns, regulatory conditions, and building use, to generate programmatic and morphological typologies for urban intensification, aligned with the city's existing urban policies.
- Integrated multiple geospatial datasets in QGIS to map spatial relationships between buildings, zoning conditions, and social context at the neighborhood scale.

SKILLS

- ESG & Sustainability: CSRD, Double Materiality, EU Taxonomy, UNGPs, OECD Guidelines, HRDD, HRIA, LCA (Life Cycle Assessment)
- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML
- Real Estate & Urban: Real estate feasibility modeling, Zoning and land use regulatory analysis, Urban morphology and spatial analysis
- Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)